Introduction to Machine Learning

Chapter 1: Supervised Learning

Supervised learning is a type of machine learning where the algorithm is trained on labeled data. The model learns to map inputs to outputs based on example input-output pairs.

1.1 Classification

Classification is a supervised learning task where the output is a discrete label. Common algorithms include Decision Trees, Random Forests, Support Vector Machines, and Neural Networks. The goal is to correctly categorize new, unseen examples.

1.2 Regression

Regression predicts continuous values. Linear regression models the relationship between variables using a line. The loss function is typically Mean Squared Error (MSE). Formula: MSE = (1/n) * sum((y_true - y_pred)^2)

Chapter 2: Unsupervised Learning

Unsupervised learning finds hidden patterns in data without labeled examples. It is used for clustering, dimensionality reduction, and anomaly detection.

2.1 K-Means Clustering

K-means is an iterative algorithm that partitions n observations into k clusters. Each observation belongs to the cluster with the nearest mean. Algorithm: 1) Initialize k centroids randomly 2) Assign each point to nearest centroid 3) Recompute centroids 4) Repeat until convergence.

Chapter 3: Neural Networks

Neural networks are inspired by biological neurons. A perceptron is the simplest unit. Deep learning uses many layers. Backpropagation computes gradients for training.

3.1 Activation Functions

Activation functions introduce non-linearity. Common ones: ReLU: f(x) = max(0,x); Sigmoid: f(x) = 1/(1+e^-x); Tanh: f(x) = (e^x - e^-x)/(e^x + e^-x). ReLU is preferred in hidden layers to avoid vanishing gradients.